Updates on the two polymath projects

Terence Tao · 15 June 2012

https://terrytao.wordpress.com/2012/06/15/updates-on-the-two-polymath-projects/

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Two quick updates with regards to polymath projects. Firstly, given the poll on starting the mini-polymath4 project, I will start the project at Thu July 12 2012 UTC 22:00. As usual, the main research thread on this project will be held at the polymath blog, with the discussion thread hosted separately on this blog.

Second, the Polymath7 project, which seeks to establish the "hot spots conjecture" for acute-angled triangles, has made a fair amount of progress so far; for instance, the first part of the conjecture (asserting that the second Neumann eigenfunction of an acute non-equilateral triangle is simple) is now solved, and the second part (asserting that the "hot spots" (i.e. extrema) of that second eigenfunction lie on the boundary of the triangle) has been solved in a number of special cases (such as the isosceles case). It's been quite an active discussion in the last week or so, with almost 200 comments across two threads (and a third thread freshly opened up just now). While the problem is still not completely solved, I feel optimistic that it should fall within the next few weeks (if nothing else, it seems that the problem is now at least amenable to a brute force numerical attack, though personally I would prefer to see a more conceptual solution).

Comments (2)

Luis R. Guzman, Jr. · 17 June 2012

Reblogged this on Guzman's Mathematics Weblog.

Polymath project: social problem-solving | Civil Statistician · 12 July 2012

[…] will be hosting a fourth “mini-polymath” tonight (July 12, 2012), starting at UTC 22:00, which is 6pm EST for us here on the US East Coast. If you […]